# Context-Aware Phrase Retrieval Framework for Occurrence-Level Disambiguation

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## Abstract

Phrase retrieval is a fundamental component of information retrieval, where identifying relevant text spans is essential for downstream reasoning. However, when identical phrases occur multiple times within a document, retrieving the intended occurrence remains challenging because existing methods primarily match phrase content rather than distinguish occurrences by their contextual relevance. In this paper, we propose a context-aware phrase retrieval framework that addresses this limitation by formulating retrieval as an occurrence-level task. The framework combines candidate spans from transformer-based extractive question-answering models through location-aware aggregation and large language model (LLM)-based contextual reranking. We evaluate the framework on the Phrase Retrieval (PR) and Phrase Sense Disambiguation (PSD) tasks of the Phrase-in-Context (PiC) benchmark, which represent phrase retrieval without and with occurrence-level ambiguity, respectively. On PSD, the proposed framework achieves 84.80% occurrence-level exact match (EM+loc), improving by 13.70 percentage points over the best published result (71.10%) and 8.47 percentage points over our strongest single-model baseline, Longformer-base (76.33%). On PR, the framework achieves 95.58% exact match (EM), exceeding the reported human upper bound of 95.00%. These results demonstrate that the primary benefit of contextual reranking lies in resolving occurrence-level ambiguity, where identifying the correct phrase alone is insufficient to ensure retrieval of the intended occurrence.

## CCS Concepts

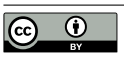
• **Information systems → Information retrieval**; • **Computing methodologies → Natural language processing**.

## Keywords

Contextual Phrase Retrieval, Phrase Sense Disambiguation, Extractive Question Answering, Large Language Models, Contextual Reranking

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## 1 Introduction

Retrieving the correct text span from a document is a fundamental problem in information retrieval, providing the supporting evidence required by downstream applications such as open-domain question answering [8, 11], fact verification [32], and retrieval-augmented generation [18]. Although modern retrieval systems have become increasingly effective at locating relevant text spans [9, 12, 14, 36], they primarily focus on identifying the appropriate text content rather than determining which specific occurrence is intended within a document. Consequently, when identical phrases appear multiple times with different contextual meanings [21], retrieving an incorrect occurrence despite matching the target text can still lead to retrieval failures and incorrect downstream reasoning [7].

Pham *et al.* [24] formalized this challenge through the Phrase-in-Context (PiC) benchmark, which comprises two complementary tasks: Phrase Retrieval (PR), where retrieving the correct phrase is generally sufficient, and Phrase Sense Disambiguation (PSD), where a query phrase may correspond to multiple identical occurrences within the same document, only one of which expresses the intended meaning. Unlike PR, PSD requires models to distinguish between candidate occurrences using contextual semantics rather than surface-level phrase matching.

Existing state-of-the-art approaches formulate both tasks using the transformer-based extractive question-answering paradigm, where a model predicts a single best-scoring span for a given query. While this formulation performs well on PR, it is inherently limited for PSD. Individual models may retrieve complementary candidate occurrences, yet retaining only the highest-ranked prediction can discard correct occurrences ranked lower by other models. Moreover, combining predictions from multiple models frequently produces duplicate or boundary-inconsistent spans corresponding to the same occurrence, resulting in a cluttered candidate space that is unsuitable for effective occurrence-level contextual reasoning.

Table 1 illustrates this challenge. The phrase "close embrace" appears twice in the same document: once describing a senator's political alignment and once referring to a partner-dance hold. Despite being strong extractive QA models trained for phrase retrieval, all three models assign near-maximal prediction probabilities to an incorrect occurrence when evaluated on PSD. This example demonstrates that the challenge is neither model uncertainty nor disagreement, but the inability of extractive span prediction to determine

which occurrence best matches the intended contextual meaning of the query. Therefore, simply combining predictions from multiple extractive QA models is insufficient to resolve occurrence-level ambiguity.

**Table 1: An instance from the PiC: PSD task. "Close embrace" occurs twice with different senses; all three extractive QA models confidently retrieve the incorrect occurrence.**

| |
|---|
| **Query phrase:** "intimate contact" |
| **Occurrence 1 (LOC-1):** "…his close embrace of Trump, reversing his outspoken criticism of Trump in the 2016 campaign…" |
| **Occurrence 2 (LOC-2):** "When in close embrace, the dance is led (and followed) with the whole body…" |
| **ELECTRA prediction:** Occurrence 1 (conf. 0.999999) |
| **RoBERTa prediction:** Occurrence 1 (conf. 0.999999) |
| **Longformer prediction:** Occurrence 1 (conf. 0.999999) |
| **Gold answer:** Occurrence 2 |

Motivated by this limitation, we propose a context-aware phrase retrieval framework that explicitly addresses occurrence-level ambiguity through a three-stage pipeline. First, an ensemble of extractive QA models generates multiple candidate occurrences instead of committing to a single prediction. Next, a location-aware aggregation module consolidates duplicate and boundary-inconsistent predictions into a compact set of distinct candidate occurrences. Finally, an LLM performs contextual reranking only when multiple candidate occurrences remain, selecting the occurrence whose contextual meaning best matches the query while avoiding unnecessary reasoning for unambiguous cases.

To summarize, the contributions of this paper are as follows:

- We introduce a context-aware phrase retrieval framework for occurrence-level disambiguation, which reformulates phrase retrieval as an occurrence-level retrieval problem by integrating ensemble-based extractive QA candidate generation, location-aware candidate aggregation, and LLM-based contextual reranking.
- We present a comprehensive evaluation on the PiC: PR and PSD benchmarks, comparing ranking-based embedding similarity methods and span-selection (QA-based) baselines under both zero-shot and task-specific fine-tuning settings. Results demonstrate that the proposed framework substantially improves occurrence-level performance on PSD while achieving competitive performance on the near-saturated PR task.
- We conduct extensive experimental analyses, including component ablation, oracle candidate-recall evaluation, reranker comparison, and error taxonomy analysis, to investigate the effects of ensemble diversity, candidate aggregation, and reranking strategies on occurrence-level phrase retrieval performance.

## 2 Related work

### 2.1 Phrase similarity and contextual phrase retrieval

Early research on phrase-level semantic similarity primarily focused on context-free evaluation, using resources such as PPDB-annotated datasets [23], PPDB-filtered and PAWS-short [34, 38], BiRD [1], and Turney's compositional similarity dataset [33]. Similarly, Wieting et al. [35] learned phrase representations from PPDB paraphrase pairs. However, these benchmarks mainly evaluate lexical or paraphrastic similarity and often contain substantial surface-level overlap between phrase pairs, reducing the need for contextual interpretation [24]. To address this limitation, WiC [25] introduced contextual word-level semantic evaluation by requiring models to determine whether a word has the same meaning across different contexts. This idea was later extended to multi-word expressions through the Phrase-in-Context (PiC) benchmark [24], which introduced Phrase Similarity (PS), Phrase Retrieval (PR), and Phrase Sense Disambiguation (PSD) tasks. While existing systems achieve strong performance on PR, PSD remains challenging because the correct retrieval target depends not only on phrase content but also on identifying the occurrence whose surrounding context matches the intended meaning.

Our work builds upon this observation by addressing phrase retrieval at the occurrence level. Instead of treating retrieval as selecting a single matching span, we generate and consolidate multiple candidate occurrences before applying contextual disambiguation.

### 2.2 Extractive question answering for span retrieval

Extractive question answering has established the dominant paradigm for retrieving text spans from documents, where models predict the start and end positions of an answer span given a query. Transformer-based architectures such as BERT [4] introduced contextual span prediction, followed by improved pretrained models including RoBERTa [19], ELECTRA [3], ALBERT [16], and XLNet [37]. SpanBERT [10] further optimized transformer representations for span-level understanding, while Longformer [2] enabled efficient processing of long documents through sparse attention mechanisms. Beyond general question answering, several studies have explored phrase-oriented retrieval and representation learning. DensePhrase [17] investigated dense phrase retrieval, while PhraseBERT [34] specialized transformer representations for phrase-level semantic similarity. Ensemble methods have also been widely studied to improve prediction robustness by combining complementary models [5].

Although these approaches are effective at identifying relevant spans, they typically prioritize the highest-scoring prediction and are not designed to explicitly reason over multiple occurrences of the same phrase. In contrast, our framework leverages multiple extractive QA models to generate diverse candidate occurrences and performs subsequent aggregation and contextual disambiguation.

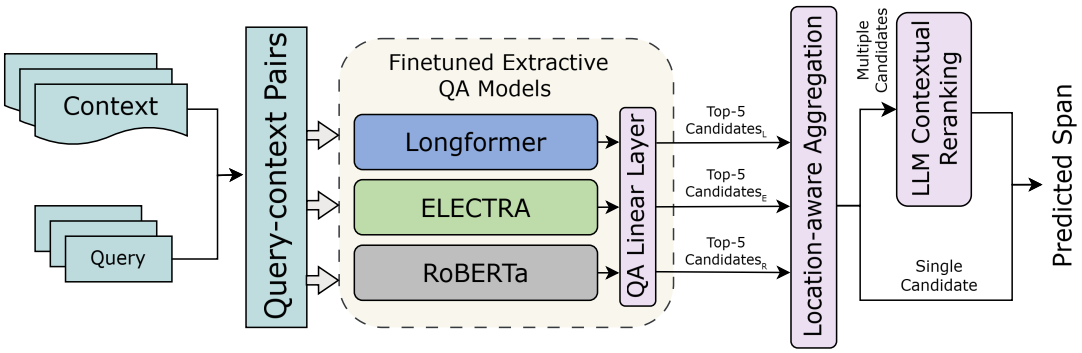



**Figure 1: Overview of the proposed context-aware phrase retrieval framework.**

## 2.3 LLM-based reranking and contextual disambiguation

Recent information retrieval research has increasingly explored large language models as contextual rerankers. RankGPT [30] demonstrated that LLMs can perform zero-shot listwise passage reranking, motivating subsequent approaches such as RankVicuna [26] and RankZephyr [27], which transfer ranking capabilities to smaller open models. Other studies investigated pairwise and setwise prompting strategies to improve ranking robustness and reduce sensitivity to candidate ordering [28, 39]. Beyond retrieval ranking, LLMs have also shown promise in contextual semantic reasoning tasks, including word sense disambiguation [15, 20]. These studies demonstrate the ability of LLMs to interpret meaning from surrounding context, making them suitable for resolving ambiguity that remains after conventional retrieval.

However, directly applying LLMs to span retrieval remains challenging due to the difficulty of producing precise span locations. Therefore, our framework employs LLMs only as contextual rerankers after candidate extraction and aggregation, allowing them to focus on selecting among plausible phrase occurrences rather than performing exact span retrieval.

# 3 Proposed framework

Our proposed context-aware phrase retrieval framework addresses occurrence-level phrase retrieval through progressive candidate refinement and contextual disambiguation. It consists of three sequential stages: (1) candidate generation using transformer-based span-selection models, (2) location-aware candidate aggregation, and (3) LLM-based contextual reranking. The overall architecture of the proposed framework is illustrated in Fig. 1.

## 3.1 Candidate generation

Candidate phrase occurrences are generated using an ensemble of three transformer-based extractive question-answering models: Longformer, ELECTRA, and RoBERTa. We selected these models to leverage complementary pretrained representations: Longformer employs sparse attention for long-context processing, ELECTRA uses a discriminative pretraining objective, and RoBERTa uses optimized masked language modeling. Their complementary candidate predictions are evaluated in Section 4.3.2. Each model is fine-tuned for phrase retrieval following the standard extractive question-answering paradigm [24] and independently predicts candidate spans for a given context document and query phrase.

The query phrase and context document are first tokenized and encoded using the corresponding tokenizer of each model. For documents exceeding the maximum input length, an overlapping sliding-window strategy is employed to ensure complete context coverage. The encoded inputs are then processed by the standard extractive question-answering span prediction head to estimate start and end logits for candidate phrase occurrences. During post-processing, each valid span is scored by summing its start and end logits. A softmax over the 20 highest-scoring valid spans converts these scores into probabilities. Rather than retaining only the highest-scoring prediction, we preserve the top-$k$ candidate spans ($k = 5$) from each model, together with their probabilities and character offsets, since the correct phrase occurrence may not be ranked first by an individual model. The resulting candidate sets are subsequently forwarded to the candidate aggregation stage.

## 3.2 Location-aware candidate aggregation

Since different span-selection models often retrieve complementary phrase occurrences, the candidate lists produced by all models are merged to form a unified candidate set for each retrieval instance. First, a phrase-plausibility filter removes unlikely candidates, including spans exceeding six words or crossing sentence boundaries. Exact duplicate candidates are then removed based on their phrase text and character offsets while recording the number of supporting models together with their maximum and average prediction probabilities.

Although different models frequently retrieve the same phrase occurrence, slight variations in predicted span boundaries often produce overlapping candidates corresponding to the same underlying occurrence. To eliminate these boundary inconsistencies, overlapping candidates are clustered according to their character offsets, and each cluster is represented by the candidate with the highest average prediction probability [22]. This location-aware aggregation process reduces redundant boundary variants while preserving unique phrase occurrences, yielding a compact candidate set for subsequent contextual reranking.

**LLM occurrence-level reranking prompt, shown with a real test instance**

**System:** `You are an expert in lexical semantics and phrase sense disambiguation.`

**User:** `You will be given a document and a query phrase that represents the target contextual meaning. Candidate occurrences are marked in the document with location tags ([LOC-1]...[/LOC-1]) and listed below with their exact text and location. Candidates may share identical or near-identical text; judge by surrounding context and meaning, not text alone. Candidate letters are assigned randomly and carry no significance.`

`Document:` *"...Mizoram is a stronghold for Mrs. Hume's pheasant (*Syrmaticus humiae*). There is also a [LOC-1]rare record[/LOC-1] of the wild water buffalo from the state... Take Aim is a studio album by American hard bop tenor saxophonist Harold Land. The album was recorded in Los Angeles in 1960 but released only in 1980 via [LOC-2]Blue Note[/LOC-2] label. This [LOC-3]rare record[/LOC-3] was re-released on CD in 2005 and 2012..."*

`Query phrase: "`*limited note*`"`

`Candidates:`
*B: "rare record" (at LOC-1)*
*C: "Blue Note" (at LOC-2)*
*A: "rare record" (at LOC-3)*

`Task: Choose the candidate whose contextual meaning best matches the sense of the query phrase. Output exactly one capital letter corresponding to your chosen candidate (e.g., A).`

`Answer: B`

**Figure 2: Prompt for LLM-based occurrence-level contextual reranking, instantiated on a real PSD test case (id 13287-1). *Blue italics* mark the per-instance fields: the source document, with each deduplicated candidate span wrapped in a unique `[LOC-i]` tag; the target query phrase; and each candidate's text and location, listed under a randomly permuted letter to remove positional bias. "Rare record" appears twice here with identical text but different senses, a wildlife sighting vs. a vinyl reissue, alongside the decoy "Blue Note". The model must rely on context rather than surface text to pick candidate B, matching gold.**

### 3.3 LLM-based contextual reranking

Following location-aware candidate aggregation, the remaining candidates represent distinct phrase occurrences requiring semantic disambiguation. Retrieval instances containing a single candidate are resolved directly, whereas instances with multiple candidates are forwarded to the LLM for contextual reranking. For each instance, the query phrase, the original context document, and the aggregated candidate occurrences are formatted using the prompt shown in Fig. 2 and provided as input to Gemini 3.5 Flash [31]. Using deterministic decoding, the model selects the candidate occurrence whose contextual meaning best matches the query phrase, and the selected occurrence is returned as the final retrieval result.

## 4 Experiments and evaluation

### 4.1 Dataset

We evaluate the proposed framework on the Phrase Retrieval (PR) and Phrase Sense Disambiguation (PSD) tasks of the Phrase-in-Context (PiC) benchmark [24], a large-scale benchmark designed to evaluate phrase understanding under contextual semantics.

**Phrase Retrieval (PR)** PR is a semantic retrieval task in which a model is given a query phrase and a document containing a semantically equivalent target phrase, and must identify the correct text span corresponding to the query. The benchmark provides two retrieval settings: *PR-pass*, where retrieval is performed over a short passage, and *PR-page*, where the search space is an entire Wikipedia page. Following prior work [24], we evaluate on the *PR-pass* benchmark, which contains 28147 instances divided into 20147 training, 3000 validation, and 5000 test examples.

**Phrase Sense Disambiguation (PSD)** PSD extends phrase retrieval by introducing occurrence-level ambiguity. Each example consists of a two-paragraph document in which the same target phrase appears exactly twice with different meanings. Given a query phrase expressing one intended sense, the model must retrieve the occurrence whose contextual meaning matches the query. The final PSD benchmark contains 4,858 examples after manual verification and quality filtering [24]. Following the evaluation protocol used in prior work, we use 1358 examples for training, 500 for validation, and 3000 for testing. Compared with PR, PSD requires contextual reasoning to distinguish between multiple identical phrase occurrences rather than simply retrieving the correct phrase text.

### 4.2 Experimental setup

All experiments are implemented using PyTorch Lightning [6] and executed on the Lightning AI platform, utilizing GPU-accelerated tensor computation. Fixed train, val, and test splits (~20K/3K/5K for PR and ~2K/0.5K/3K for PSD) are used to ensure fair comparison with previous work. In addition, LLM inference is performed using Google Cloud Batch Inference[1].

For phrase candidate generation, multiple pre-trained transformer models, including Longformer-base[2], ELECTRA-large[3], and RoBERTa-large[4], are fine-tuned sequentially on PR and PSD following the standard extractive question answering framework proposed in the original PiC benchmark [24]. Models are trained for

[1] https://cloud.google.com/vertex-ai
[2] https://huggingface.co/allenai/longformer-base-4096
[3] https://huggingface.co/google/electra-large-discriminator
[4] https://huggingface.co/FacebookAI/roberta-large

**Table 2: Comparison of prior work, human upper bound, additional extractive QA baselines, and the proposed framework on PR and PSD. For PSD, both text-based (EM/F1) and occurrence-level (EM+loc/F1+loc) performance are reported; PR is evaluated at the text level and reports EM/F1 only. Prior-work results are taken from the original PiC benchmark [24] and cover ranking-based and QA/span-selection (SS) approaches under different training settings. Additional baselines comprise three individual extractive QA models (ELECTRA-large, Longformer-base, and RoBERTa-large).**

| Task | Baseline type | System (setting) | EM | F1 | EM+loc | F1+loc |
|---|---|---|---|---|---|---|
| PSD | Ranking | Not evaluated[a] | – | – | – | – |
| | QA/SS (zero-shot) | Longformer-base (PR-pass-trained) | 96.17 | 96.88 | 62.72 | 62.83 |
| | | Longformer-large (PR-pass-trained) | 96.32 | 96.91 | 59.72 | 59.82 |
| | QA/SS (fine-tuned) | Longformer-base (PR-pass + PSD∼2K) | – | – | 71.10 | 71.30 |
| | | Longformer-large (PSD∼2K only) | – | – | 66.27 | 67.10 |
| | *Additional baselines* | | | | | |
| | QA/SS (PR-pass + PSD∼2K trained) | ELECTRA-large | 95.30 | 96.24 | 75.40 | 75.59 |
| | | Longformer-base (reimplemented) | 95.47 | 96.20 | 76.33 | 76.49 |
| | | RoBERTa-large | 96.00 | 96.76 | 74.93 | 75.06 |
| | – | Human upper bound | – | – | 95.00 | – |
| | **Proposed framework** | | **95.50** | **96.34** | **84.80** | **84.97** |
| PR | Ranking (untrained) | BERT | 47.44[b] | – | – | – |
| | | BERT-large | 42.80[b] | – | – | – |
| | Ranking (PR-trained) | PhraseBERT | 59.02[b] | – | – | – |
| | | SentenceBERT | 53.14[b] | – | – | – |
| | QA/SS (PR-pass-trained) | Longformer-large | 94.28 | 95.53 | – | – |
| | | Longformer-base | 94.26 | 95.58 | – | – |
| | *Additional baselines* | | | | | |
| | QA/SS (PR-pass-trained) | ELECTRA-large | 96.06 | 97.16 | – | – |
| | | Longformer-base (reimplemented) | 93.52 | 94.76 | – | – |
| | | RoBERTa-large | 95.30 | 96.50 | – | – |
| | – | Human upper bound | 95.00 | – | – | – |
| | **Proposed framework** | | **95.58** | **96.82** | – | – |

[a] Ranking-based models were not evaluated on PSD in the original benchmark, since they already performed substantially worse than span-selection models on PR-pass [24].
[b] For ranking-based baselines, EM corresponds to Top-1 retrieval accuracy over enumerated candidate spans, following the original benchmark's Table 4; this differs procedurally from token-level EM computed for span-selection models.

two epochs using the Adam optimizer [13] with a learning rate of $1 \times 10^{-4}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 1 \times 10^{-8}$. The maximum sequence length and document stride are set to 512 and 128 tokens, respectively. Mixed-precision (FP16) training is enabled, and all experiments use a fixed random seed of 42. During inference, each model retains its top-$k = 5$ predictions, which are merged using the proposed location-aware aggregation strategy described in Section 3.2. For instances that remain ambiguous after candidate aggregation, Gemini 3.5 Flash is employed as a contextual reranker to identify the occurrence whose meaning best matches the query phrase in context. The model is configured with a temperature of 0.1, top-$p$ of 0.95, and thinking budget set to -1.

Performance is evaluated using text-only and location-aware Exact Match (EM, EM+loc) [29], token-level F1 (F1, F1+loc) [29], and oracle candidate recall. EM and F1 measure text-level correctness based on phrase matching and token overlap, respectively, while their location-aware variants additionally require the predicted occurrence to match the gold character offset.

## 4.3 Results and analysis

*4.3.1 Overall retrieval performance.* **Q:** *How does the proposed framework compare with existing phrase retrieval approaches on PR and PSD?*

Previous work on the PiC benchmark has explored two fundamentally different retrieval paradigms. *Ranking-based* methods encode the query and candidate phrases into dense semantic representations and retrieve the candidate with the highest embedding similarity. In contrast, *extractive span-selection (QA-based)* methods fine-tune pretrained language models to directly predict the start and end positions of the target span within a passage. For PSD, QA-based models have been evaluated both in a zero-shot setting (trained only on PR-pass) and after additional fine-tuning on a PSD training split.

Table 2 compares the proposed framework with previously published retrieval methods, our three evaluated extractive QA models, and the human performance reported in the original PiC benchmark [24].

**Table 3: Configuration-wise retrieval performance (%). For PSD, both text-level (EM/F1) and occurrence-level (EM+loc/F1+loc) metrics are reported; PR is evaluated using text-level EM/F1 only. Individual models are evaluated using their top-ranked candidate (Top-1). Top-1 vote selection combines the top-ranked candidates from all three retrievers, merging exact duplicates. Candidate pooling combines their top-five candidates and merges exact duplicates. Location-aware aggregation further consolidates overlapping span variants, as described in subsection 3.2. For all three configurations, the candidate receiving the most model votes is selected, with maximum prediction probability breaking ties. Finally, LLM contextual reranking uses Gemini 3.5 Flash to select the intended occurrence from the aggregated candidates; instances with a single remaining candidate are resolved directly.**

| Task | Configuration | EM | F1 | EM+loc | F1+loc |
|---|---|---|---|---|---|
| PSD | ELECTRA-large (Top-1) | 95.30 | 96.24 | 75.40 | 75.59 |
| | Longformer-base (Top-1) | 95.47 | 96.20 | 76.33 | 76.49 |
| | RoBERTa-large (Top-1) | 96.00 | 96.76 | 74.93 | 75.06 |
| | Top-1 vote selection | 97.17 | 97.72 | 79.60 | 79.73 |
| | Candidate pooling (Top-5) | 97.40 | 97.93 | 73.13 | 73.26 |
| | Location-aware aggregation | 97.60 | 98.19 | 80.50 | 80.61 |
| | LLM reranking (Proposed framework) | 95.50 | 96.34 | 84.80 | 84.97 |
| PR | ELECTRA-large (Top-1) | 96.06 | 97.16 | – | – |
| | Longformer-base (Top-1) | 93.52 | 94.76 | – | – |
| | RoBERTa-large (Top-1) | 95.30 | 96.50 | – | – |
| | Top-1 vote selection | 96.62 | 97.54 | – | – |
| | Candidate pooling (Top-5) | 96.54 | 97.55 | – | – |
| | Location-aware aggregation | 96.36 | 97.42 | – | – |
| | LLM reranking (Proposed framework) | 95.58 | 96.82 | – | – |

The results highlight a clear contrast between the PR and PSD benchmarks. On PR, the proposed framework achieves 95.58% EM, outperforming the strongest published baseline, Longformer-large (94.28%), by 1.30 percentage points. This score also slightly exceeds the reported human performance of 95.00%. However, ELECTRA-large, evaluated as an additional extractive QA baseline, achieves a higher EM of 96.06%. These results suggest that text-level phrase retrieval leaves limited room for improvement even with targeted contextual retrieving mechanisms, where each semantic meaning corresponds to a unique occurrence.

PSD presents a fundamentally different picture. Although previous QA-based retrievers achieve approximately 96% text-based EM, occurrence-level performance remains substantially lower, reaching only 71.10% EM+loc after task-specific fine-tuning. The proposed framework improves this to 84.80% EM+loc, a gain of 13.70 percentage points over the best published result and 8.47 percentage points over our strongest reimplemented single-model baseline, Longformer-base (76.33% EM+loc). The gains indicate that the principal challenge is no longer retrieving the correct phrase, but identifying the intended occurrence when multiple contextually plausible matches exist.

Taken together, these findings emphasize that the effectiveness of contextual reranking depends on the retrieval setting. When occurrence ambiguity is absent, as in PR, additional contextual reasoning provides only limited benefit because the correct phrase is typically retrieved directly. In contrast, PSD requires reasoning beyond phrase matching, making occurrence-level contextual disambiguation the primary factor driving performance gains. The following sections analyze how each component of the proposed framework contributes to these improvements.

*4.3.2 Component analysis.* **Q:** *How does each component of the proposed framework contribute to the overall retrieval performance?*

To isolate the contribution of each component, we evaluate progressively expanded configurations, beginning with individual extractive QA retrievers, followed by Top-1 vote-based selection, Top-5 candidate pooling, location-aware aggregation, and finally contextual reranking. Table 3 summarizes each configuration and their performance on both PSD and PR.

**Table 4: Prediction changes relative to the final framework. ΔPrimary denotes the improvement in the primary evaluation metric (EM+loc for PSD and EM for PR). *Fixed* counts instances where the component's incorrect prediction is corrected by the proposed framework, whereas *Broke* counts instances where the component's correct prediction becomes incorrect.**

| Task | Component | ΔPrimary | Fixed | Broke |
|---|---|---|---|---|
| PSD | ELECTRA-large | +9.40 | 473 | 191 |
| | Longformer-base | +8.47 | 482 | 228 |
| | RoBERTa-large | +9.87 | 501 | 205 |
| | Location-aware aggregation | +4.30 | 337 | 208 |
| PR | ELECTRA-large | −0.48 | 93 | 117 |
| | Longformer-base | +2.06 | 205 | 102 |
| | RoBERTa-large | +0.28 | 116 | 102 |
| | Location-aware aggregation | −0.78 | 62 | 101 |

The effect of each component differs substantially between the two benchmarks. On PSD, the strongest individual retriever, Longformer, achieves 76.33% EM+loc. Combining the models' Top-1

**Table 5: Comparison of rerankers on 563 PSD test instances with identical candidate sets across all runs. The aggregation-based selection chooses the aggregated candidate supported by the most models, with ties broken by maximum probability, as in Table 3. *Correct* denotes the number of instances with the correct occurrence-level prediction. *Fixed* and *Broke* denote predictions corrected and errors introduced relative to the aggregation-based selection; *Net* is their difference. Runtime (Min./1k) is normalized batch inference time per 1,000 instances.**

| Method | Correct | EM+loc (%) | Fixed | Broke | Net | Min./1k |
|---|---|---|---|---|---|---|
| Location-aware aggregation (no LLM reranker) | 417 | 74.1 | – | – | – | – |
| Gemini 2.5 Flash | 436 | 77.4 | 78 | 59 | +19 | 3.30 |
| Gemini 2.5 Pro | 447 | 79.4 | 87 | 57 | +30 | 56.40 |
| Gemini 3.5 Flash | 463 | 82.2 | 91 | 45 | +46 | 7.20 |

predictions through vote-based selection raises this to 79.60%, an improvement of 3.27 percentage points that demonstrates the benefit of model diversity. However, pooling all three models' Top-5 candidates without clustering reduces EM+loc to 73.13%. Overlapping spans with slightly different boundaries can represent the same occurrence as separate candidates, fragmenting support for that occurrence. Location-aware aggregation addresses this problem, raising EM+loc to 80.50%, a gain of 7.37 percentage points over the unclustered Top-5 candidate pool and 0.90 points over Top-1 Vote Selection. Contextual reranking provides a further improvement to 84.80%, yielding an overall gain of 8.47 percentage points over the strongest individual retriever. In contrast, text-only EM decreases from 97.60% after aggregation to 95.50% after reranking. This indicates that reranking improves occurrence selection even though it can introduce phrase-text or span-boundary errors.

A different pattern emerges on PR. Since PR is evaluated at the phrase-text level without requiring occurrence-level disambiguation, aggregation does not improve EM over Top-1 Vote Selection (96.36% vs. 96.62%). Adding contextual reranking further reduces EM to 95.58%, suggesting that explicit occurrence-level reasoning is most valuable when multiple candidate occurrences compete for the same query rather than when retrieval itself is already unambiguous.

To better understand these changes, Table 4 compares the final framework's predictions with those of each individual retriever and the aggregation stage.

On PSD, the final framework corrects substantially more instances than it degrades relative to each individual retriever. Compared with location-aware aggregation, reranking fixes 337 previously incorrect occurrences but introduces 208 new errors, resulting in a net gain of 129 correct predictions. This is consistent with the 4.30-percentage-point improvement in EM+loc.

The opposite trend is observed on PR. Relative to location-aware aggregation, reranking fixes 62 instances but changes 101 previously correct predictions into incorrect ones, resulting in a net loss of 39 correct predictions. This corresponds to the decrease in EM from 96.36% to 95.58%. These results suggest that contextual reranking is better viewed as a targeted approach to occurrence-level disambiguation than as a general post-processing step for phrase retrieval.

The complementary behavior of the three extractive retrievers further explains the benefit of candidate expansion. On PSD, the union of their Top-5 candidate lists contains the gold occurrence in 2,973 of 3,000 instances, and all three models recover it in 2,540 instances. In another 111 instances, only one model retrieves the gold occurrence: 62 contributions come from Longformer, 25 from RoBERTa, and 24 from ELECTRA. On PR, 44 instances receive a unique contribution from one model: 15 from ELECTRA, 15 from RoBERTa, and 14 from Longformer. These results show that model diversity expands candidate coverage, while the stage-wise analysis demonstrates that effective aggregation is necessary to translate this coverage into improved occurrence-level retrieval.

*Reranker comparison.* To examine how reranker choice affects occurrence-level retrieval, Table 5 compares three Gemini models and aggregation-based selection on 563 PSD test instances with identical candidate sets across all runs. Gemini 3.5 Flash achieves the highest EM+loc (82.2%), followed by Gemini 2.5 Pro (79.4%) and Gemini 2.5 Flash (77.4%), while the aggregation-based selection achieves 74.1% on the same subset.

Paired exact McNemar tests with Holm correction across the six pairwise comparisons show that Gemini 3.5 Flash significantly outperforms the aggregation-based selection (adjusted $p = 0.0006$) and Gemini 2.5 Flash (adjusted $p = 0.012$). The remaining four pairwise differences are not statistically significant after correction. Although Gemini 2.5 Pro achieves competitive performance, its substantially higher inference time (56.40 vs. 7.20 minutes per 1,000 instances) and the absence of a statistically significant difference from Gemini 3.5 Flash favor Gemini 3.5 Flash as the final reranker, offering the best observed accuracy with lower inference cost.

#### 4.3.3 *Bottleneck analysis.* **Q:** *Is the remaining performance primarily limited by candidate generation or by occurrence disambiguation?*

Although the proposed framework substantially improves occurrence level retrieval, the remaining errors may still originate from two different stages: candidate generation or occurrence disambiguation. To identify the dominant bottleneck, we first measure the oracle recall of the candidate pool, which indicates whether the gold occurrence is present among the generated candidates regardless of the final prediction. This represents the maximum performance achievable by the reranking stage given the available candidates.

Table 6 shows that candidate generation is already highly effective. The union of the three retrievers' Top-5 lists (candidate pooling) contains the gold occurrence (exact character offsets) in 99.10% of PSD instances, compared with 94.90% for the best single model, demonstrating the complementary nature of their candidate sets.

**Table 6: Oracle recall (%) of the candidate pool at different stages of the PSD pipeline. Text-only recall ignores the offsets; location-aware recall requires the gold character offsets and the same lowercased text.**

| Stage | Text-only | Location-aware |
|---|---|---|
| Longformer-base | 98.50 | 94.90 |
| ELECTRA-large | 98.53 | 91.60 |
| RoBERTa-large | 98.33 | 92.67 |
| Candidate pooling (Top-5) | 99.83 | 99.10 |
| Location-aware aggregation | 99.50 | 98.63 |
| Proposed framework | 95.50 | 84.80 |

After location-aware aggregation, oracle recall decreases by only 0.47 percentage points, to 98.63%. Of the 41 instances (1.4%) in which the gold occurrence is absent after aggregation, 28 were never proposed by any model, while the remaining cases involve clustering choices that produce a representative span slightly shorter or longer than the gold annotation.

The small loss in oracle recall during aggregation suggests that candidate generation is not the main bottleneck. Instead, the gap between the post-aggregation oracle recall (98.63%) and the final EM+loc (84.80%) indicates that the main challenge lies in selecting the correct occurrence from the available candidates. To better understand the sources of this performance gap, Table 7 categorizes the remaining errors by type and identifies the stage of the pipeline responsible for each error.

**Table 7: Error taxonomy of the proposed framework.**

| Stage | Error category | PSD | PR |
|---|---|---|---|
| Candidate generation | Candidate missing | 41 | 88 |
| Reranking | Occurrence confusion | 302 | 43 |
| Reranking | Phrase mismatch | 81 | 88 |
| Annotation | Boundary mismatch | 30 | 58 |
| Infrastructure | Invalid LLM output | 2 | 2 |
| | **Total** | 456 | 279 |

The error distribution supports the same conclusion. On PSD, occurrence confusion is by far the most common failure mode, accounting for 302 of the 456 remaining errors. In these cases, the correct occurrence has already been retrieved, but the reranker selects a different occurrence of the same or a semantically related phrase. By comparison, only 41 errors result from missing candidates, confirming that the extractive QA ensemble already retrieves the correct occurrence for the vast majority of instances.

The pattern is noticeably different on PR. Because occurrence ambiguity is limited, occurrence confusion becomes relatively uncommon, and the remaining errors are distributed more evenly across phrase mismatches and annotation boundary differences. This mirrors the findings in Section 4.3.2, where contextual reranking provides only marginal benefit on PR.

Overall, the candidate generation stage already operates close to its practical upper bound. The remaining performance gap is therefore dominated by occurrence-level reasoning rather than candidate retrieval, suggesting that future improvements are more likely to come from stronger contextual disambiguation than from generating additional candidates. The qualitative examples in the next section illustrate representative cases from each of these error categories.

*4.3.4 Qualitative analysis.* **Q:** *What types of occurrence-level ambiguities are successfully resolved, and where does the proposed framework still fail?*

The quantitative analyses in Sections 4.3.2 and 4.3.3 show that most remaining PSD errors arise from occurrence-level disambiguation rather than candidate generation. We therefore examine representative examples illustrating three common scenarios: successful contextual disambiguation, fine-grained semantic ambiguity, and candidate generation failure.

*Successful contextual disambiguation.* Table 8 presents a representative success case. The phrase *whole development* appears twice in the document but refers to different concepts. One occurrence describes the developmental progress of mankind, whereas the other refers to the completion of a real-estate project. Although both candidate spans are lexically identical, only the first matches the intended meaning of the query phrase *entire progression*. The reranker correctly selects the gold occurrence by comparing contextual meaning rather than relying solely on span confidence.

**Table 8: Successful contextual disambiguation. Two identical phrase occurrences correspond to different meanings, and the reranker selects the contextually appropriate occurrence.**

| |
|---|
| **Query phrase:** *entire progression* |
| **Occurrence 1 (Gold):** "...the comprehension of the whole development of mankind, even of the actual fact of becoming human." |
| **Occurrence 2:** "...the Zirca, The Clinic, Forbidden City by the Indochine Group and the whole development was completed in October 2006." |
| **Prediction:** Occurrence 1 **Correct** ✓ |

This example illustrates the primary advantage of contextual reranking. Once the correct occurrence is present in the candidate pool, occurrence-level reasoning can distinguish between semantically different uses of identical phrases that conventional phrase retrieval cannot separate.

*Failure under fine-grained semantic ambiguity.* Table 9 shows a more challenging case. The phrase *final position* refers either to a person's last professional appointment or to the final board configuration in a game. Both occurrences remain semantically plausible for the query phrase *ultimate spot*, and the reranker predicts the more common career-related interpretation instead of the specialized board-game sense.

Unlike the previous example, candidate generation succeeds by retrieving both valid occurrences. The error arises entirely during occurrence-level reasoning, consistent with the dominant error category identified in the bottleneck analysis.

**Table 9: Failure under fine-grained semantic ambiguity. Both occurrences remain contextually plausible, but the reranker selects the more common interpretation.**

| **Query phrase:** *ultimate spot* | |
|---|---|
| **Occurrence 1 (Predicted):** "...with his final position in the rabbinate in Florida." | |
| **Occurrence 2 (Gold):** "The final position (the position later used to score the game) is the position on the board at the time the players pass consecutively." | |
| **Prediction:** Occurrence 1 | **Incorrect** × |

*Candidate generation failure.* Not all errors originate from contextual reasoning. Table 10 illustrates a case where the extractive QA ensemble never retrieves the gold occurrence. Instead, all candidate spans correspond to *main entrance*, while the correct occurrence *twin pillars* is absent from the candidate pool.

**Table 10: Candidate generation failure. The gold occurrence is absent from the candidate pool, so the reranker must select between two incorrect candidates.**

| **Query phrase:** *important gateway* | |
|---|---|
| **Gold occurrence (not retrieved):** "...a portico with twin pillars sheltered the main entrance." | |
| **Occurrence 1 (Predicted):** "...a pediment with oculus sat above the main entrance , and a portico with twin pillars sheltered the main entrance." | |
| **Occurrence 2:** "...and a portico with twin pillars sheltered the main entrance . The two-story brick walls..." | |
| **Prediction:** Occurrence 1 | **Gold unavailable** |

Since the correct occurrence is unavailable before reranking, no downstream reasoning method can recover the correct prediction. These examples correspond to the oracle-gap errors discussed in Section 4.3.3 and represent the practical upper bound of the proposed pipeline.

Overall, the qualitative examples complement the quantitative findings. Successful cases demonstrate that contextual reasoning effectively resolves occurrence-level ambiguity once the correct candidate is available, while the remaining failures are primarily attributable to either fine-grained semantic ambiguity or missing candidates generated by the extractive retrievers.

## 4.4 Threats to validity

Several factors should be considered when interpreting our results. First, the evaluation is limited to the PiC benchmark, as no alternative benchmark currently provides occurrence-level annotations suitable for phrase-in-context retrieval. Consequently, the generalizability of the proposed framework to other domains, document collections, and languages remains to be established. Second, the PR benchmark contains approximately 5% annotation noise, as reported by Pham et al. [24]. This limits the measurable performance ceiling on PR and warrants caution when interpreting small performance differences near the top of the benchmark. Third, the use of closed-weight Gemini models limits reproducibility and prevents independent verification of their training data. Since these models were released after the PiC benchmark became publicly available, potential training-data contamination on the evaluated splits cannot be ruled out. Evaluating open-weight models or models with documented training-data cutoffs could help address these concerns. Finally, the proposed framework optimizes candidate generation and contextual reranking independently; alternative end-to-end training strategies may yield different performance characteristics.

## 5 Conclusion and future work

We presented a context-aware phrase retrieval framework that reformulates phrase retrieval as an occurrence-level retrieval problem by combining an ensemble of extractive QA models, location-aware candidate aggregation, and LLM-based contextual reranking. Experiments on the PiC benchmark show that the proposed framework substantially improves Phrase Sense Disambiguation (PSD) while providing only modest gains on the near-saturated Phrase Retrieval (PR) task. Component, bottleneck, and qualitative analyses further demonstrate that these improvements arise primarily from contextual occurrence-level reasoning rather than improved phrase matching alone.

More broadly, our findings suggest that occurrence-level contextual reasoning is most beneficial when multiple plausible occurrences must be distinguished, but offers limited advantage when occurrence ambiguity is absent. These findings position occurrence-level contextual reasoning as a complementary capability to conventional phrase retrieval, to be applied when occurrence ambiguity is present rather than as a universal replacement.

Future work will focus on improving candidate generation, exploring efficient open-weight rerankers, extending the framework to multilingual and low-resource settings such as Bangla, and investigating joint optimization of candidate generation and contextual reranking.

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